

SAMPREETH AVVARI

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SUMMARY

AI engineer who builds production ML systems forward-deployed, in the room with non-technical stakeholders from the C-suite to domain experts. I own ambiguous problems end-to-end, from LLM pipelines, fine-tuning, and RLHF to the full-stack apps and MLOps behind them, recently replacing a six-figure vendor quote and rebuilding in weeks what prior vendors couldn't in years.

EDUCATION

Master of Science, Computer Engineering

- New York University, NY

Aug 2023 - May 2025

CGPA: 3.84 / 4

Bachelor of Technology, Computer Science and Engineering

- Jawaharlal Nehru Technological University, India

July 2017 - June 2021

CGPA: 3.92 / 4

SKILLS

Languages: Python, TypeScript, Go, Java, SQL, JavaScript, C++, **AI / ML:** PyTorch, Transformers, TRL, QLoRA/LoRA, RLHF, RAG (pgvector/Milvus), OpenVINO, ONNX, LangChain, Vertex AI Gemini, **Backend:** FastAPI, Flask, Node.js, GraphQL, REST, Pydantic/Zod, **Frontend:** React, Next.js, Tailwind, Drizzle ORM, **Data:** PostgreSQL, MySQL, MongoDB, DynamoDB, Redis, BigQuery, Kafka, **Cloud/Infra:** GCP (Cloud Run, Cloud SQL, Eventarc, Pub/Sub, Secret Manager), AWS (S3, EC2, RDS, MSK), Docker, Kubernetes, Terraform, MLflow, Airflow, Ray, **MLOps:** CI/CD (GitHub Actions, Cloud Build, Jenkins), pytest/Vitest, OpenTelemetry, Prometheus, Grafana

Certifications: Google Gen AI Leader.

WORK EXPERIENCE

AI Engineer | Hybridge Implants, New York

Sept 2025 - Present

CBCT Scan Validator - in-house dental-CT quality classifier

(<https://sampreethavvari.github.io/posts/cbct-scan-validator>)

- Built the quality gate that catches bad dental CT scans before they reach the implant-design queue, an in house Cloud Run service that replaced a \$98K up front plus \$26K/yr vendor quote and runs under \$50/mo.
- Trained classifier on frozen DentalSegmentator (nnU-Net v2) encoder with a compact 500K-parameter head, served via OpenVINO, ~5.5s end-to-end per scan, CPU-only and scale-to-zero, with a cost tuned threshold weighting false negatives 15x false positives.
- Owned the full MLOps loop (SFTP ingest, parity-checked ETL, training, serving) and gated every push with a 20-scan holdout that caught a leaky 0.80 AUROC; shipped the honest 0.63 with a human-review queue for borderline scans.

Treatment Estimator - Next.js + Postgres quoting platform

(<https://sampreethavvari.github.io/posts/treatment-estimator>)

- Built a tool treatment coordinators use to generate patient implant-cost quotes (Next.js 16, TypeScript, Drizzle, PostgreSQL, Auth.js on Cloud Run), replacing a decade-old single-price tool that couldn't handle 3 clinics' price books and that a prior vendor failed to rebuild.
- Designed a pricing engine supporting 5 quote models (flat, tiered, tiered-zoned, price-range, per-surface) behind a guided coordinator wizard that renders a patient ready PDF estimate; 100% branch coverage with golden snapshots locking every wizard path.
- Shipped end-to-end in ~1 month, built directly with the clinical stakeholders who use it daily, running under \$35/mo.

Doc Coach - LLM consultation-QA pipeline

(<https://sampreethavvari.github.io/posts/clinical-rag>)

- Built the system that grades every implant consultation a doctor runs: it pulls the Zoom transcript, scores the doctor against a 7-criterion clinical rubric, and emails a color coded coaching report to the doctor and CEO (Cloud Run, FastAPI, Vertex AI Gemini).
- Made LLM output trustworthy with versioned rubric prompt and JSON Schema contract on every response (one retry on schema failure), cutting hallucinations ~35% vs no-schema under HIPAA-eligible controls; +130% acceptance, +43% revenue downstream.
- Solved per-meeting doctor-to-coordinator identity (participant list, transcript alias matching, LLM tie-break) and avoided Workspace Domain-Wide Delegation with a scoped user-OAuth grant.

Software Engineer | Shure Incorporated, India

Aug 2021 - Aug 2023

- Built and deployed Flask REST APIs on AWS (DynamoDB, S3, Amazon MSK) for a speech-audio analytics pipeline ingesting 500K+ daily events via Kafka, refactoring synchronous calls into async event-driven workers to cut end-to-end processing time 20%.
- Owned quality and delivery: Factory/Strategy pytest/Selenium framework (150+ tests, 90%+ coverage, 3 services) cut manual QA 70%, while Jenkins CI/CD across 3 tracks with canary deploys and rollback cut deployment errors 40% and raised velocity 35%.

RESEARCH EXPERIENCE

Machine Learning Researcher | New York University, New York (<https://sampreethavvari.github.io/posts/llama-rlhf>) May 2024 - Sep 2025

- Fine-tuned Llama 3.1 8B for persuasive counter-arguments: built Python/SQL/Airflow ETL turning 118 CMV shards into 38K preference pairs (schema enforcement, failure alerting), then trained a reward model with Transformers, TRL, QLoRA on GPU clusters.
- Ran the full RLHF lifecycle (SFT then GRPO/PPO policy optimization) with checkpointed recovery and WandB tracking of loss, precision, recall, and F1, reaching a ~67% win rate over the base model on human evaluation.

PROJECTS

Loan Radar

(<https://hi.switchy.io/cv-i>)

- Built a self-retraining loan-default scoring platform (grad project): FastAPI microservices, MLflow-tracked CI/CD with quality gates, and automated retraining on Terraform-provisioned Kubernetes (Ray), delivering benchmarked low-latency, rollback-safe predictions on one node.